Rasal Musthafa

Computer Science & Engineering Student | Robotics & Computer Vision | Full-Stack Development

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SUMMARY

Computer Science & Engineering student exploring **Robotics, Edge Comuting, Computer Vision, Machine Learning, and Full-Stack Development**. Interested in building practical projects, learning emerging technologies, and gaining experience in intelligent and autonomous systems.

EDUCATION

T.K.M College of Engineering Kollam, Kerala
B.Tech in Computer Science & Engineering | CGPA: 9.48/10.0 **2024–2028**

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Computer Architecture

EXPERIENCE

Web Master Mar 2026–Present
IEEE Circuits and Systems Society Kerala Chapter **Kerala**

- Develop and maintain web platforms for the IEEE CASS Kerala Chapter, supporting technical events, member engagement, and organizational activities.
- Build responsive interfaces and automate web workflows to streamline event and content management.

Web Team Lead Mar 2026–Present
IEEE SB TKMCE **Kollam**

- Lead the web team in developing and deploying responsive websites and digital platforms for IEEE student activities and events.
- Coordinate development workflows using Git and modern web technologies while reviewing and integrating team contributions.

Tech Head – Full Stack Developer Oct 2025–Present
Produit Academy **Startup**

- Develop full-stack applications using React.js, Next.js, Django, Node.js, and PostgreSQL.
- Design REST APIs, database schemas, authentication flows, and deployment workflows for web applications.
- Collaborate on application architecture, feature development, debugging, and code quality.

Research Intern – Computer Vision & Robotics May 20, 2026–Jun 26, 2026
AGCL Lab, Indian Institute of Technology Madras **Chennai, India**

- Worked on computer vision, visual SLAM, and 3D scene reconstruction in a research-driven robotics environment.
- Explored and implemented research frameworks including SplaTAM and MonoGS for vision-based 3D reconstruction and SLAM.
- Gained practical experience with visual perception, camera-based localization, and reconstruction pipelines for robotic systems.

Reserve Software Team Member Jun 2025–Apr 2026
DRONA TKMCE – SAE TKMCE **Kollam**

- Contributed to UAV software development for autonomous drone applications and competition systems.
- Developed computer vision pipelines using OpenCV and MediaPipe for real-time gesture recognition.
- Worked on deep-learning-based gesture classification for UAV control and search-and-rescue applications.

Network Technician Jul 2024–Sep 2024
Keralavision Broadband Limited **Kannur**

- Assisted with broadband network installation, troubleshooting, and maintenance.
- Diagnosed connectivity issues and supported field-level network operations.

PROJECTS

Autonomous Warehouse Mapping Robot | ***ROS2, Gazebo, SLAM Toolbox*** **GitHub**

- Developed an autonomous exploration and mapping pipeline for a differential-drive Romi robot using ROS2 and Gazebo.
- Implemented hybrid potential-field navigation with obstacle avoidance, novelty-based exploration, and autonomous recovery behaviors.
- Integrated LiDAR, IMU, wheel encoder, and RGB-D camera data using sensor fusion for localization and mapping.
- Generated synchronized RGB, depth, point-cloud, odometry, and ground-truth datasets for 3D reconstruction research.
- Built occupancy-grid maps using SLAM Toolbox and automated data collection for COLMAP and NeRF-based reconstruction workflows.

AI Gesture-Controlled Drone | ***MediaPipe, OpenCV, TensorFlow Lite, YOLO*** **GitHub**

- Engineered a real-time gesture recognition pipeline using MediaPipe Pose and skeletal feature extraction for drone control.
- Trained sequential deep-learning models to classify ICAO/IMO search-and-rescue gestures.
- Integrated lightweight object-detection models including EfficientDet-Lite and YOLOv3-Tiny for edge-oriented visual perception.

DonorLink – Smart Blood Bank System | ***React.js, Django, PostgreSQL*** **GitHub**

- Developed a full-stack platform connecting hospitals, organizations, and blood donors through a React frontend and Django backend.
- Implemented AI-assisted donor matching using geographic proximity and blood-group compatibility for emergency requests.
- Built role-based dashboards for donors, hospitals, and administrators with inventory and request management.
- Integrated OpenStreetMap-based location services to support geographically relevant donor matching.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, JavaScript, HTML/CSS

Frameworks & Backend: React.js, Next.js, Node.js, Django, REST APIs

Databases: PostgreSQL, Firebase Firestore, MongoDB

Robotics & Simulation: ROS2, Gazebo, SLAM Toolbox, Robot Localization, PyBullet

AI & Computer Vision: OpenCV, MediaPipe, TensorFlow Lite, YOLO, Machine Learning

Tools: Git, Linux

Core Areas: Data Structures & Algorithms, Robotics, Autonomous Navigation, SLAM, Computer Vision, Machine Learning, Full-Stack Development

CERTIFICATIONS

- Computer Architecture and Organization – NPTEL — Nov 2025
- The Joy of Computing using Python – NPTEL — May 2025
- Getting Started with Linux Fundamentals (RH104) – Red Hat — Jan 2025
- The Basics of Google Cloud Compute – Google Cloud Skills Boost — Oct 2024